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Abstract

The equilibrium quantum vacuum possesses enormous energy density but does not produce local spacetime curvature, because it is spatially homogeneous — a uniform load on an elastic membrane creates no local bending. Only inhomogeneous perturbations of the vacuum generate curvature. We show that such perturbations are mediated by the chiral condensate of quantum chromodynamics, and that this condensate has a finite response time set by the confinement scale of the strong interaction. The response time, approximately 3.3×10^{-24} seconds, is determined by the QCD scale and measured in pion-nucleon scattering experiments. If a particle exists for less than this time, the condensate does not shift, and no local gravitational effect is produced. This threshold classifies all known particles: the Z boson, W boson, and top quark are too short-lived to activate gravitational coupling; the $\Delta(1232)$ baryon and ρ meson lie at the boundary; protons, neutrons, pions, and electrons gravitate at full strength. Virtual vacuum fluctuations produce no local curvature because they are spatially uniform and individually too short-lived for the condensate to respond. The minimum action for gravitational activation equals one quarter of the Planck constant — a ratio of two measured QCD quantities (the pion-nucleon sigma term and the QCD confinement scale). The gravitational threshold is set entirely by nuclear physics, not by the Planck scale. We argue that the Planck length is not a physical boundary but a virtual focal point of dimensional analysis, lying twenty orders of magnitude below the actual threshold of gravitational coupling.

Gravitational Activation Threshold and the Irrelevance of Planck Scale: Vacuum Condensate Dynamics as the Origin of Gravitational Coupling

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Abstract

We derive a gravitational activation threshold from QCD condensate dynamics: a perturbation of energy E existing for time Δt produces spacetime curvature only if the chiral condensate has sufficient time to shift in response. The condensate response time $\tau_{LK} = \hbar/\Lambda_{QCD} \approx 3.3 \times 10^{-24}$ s defines the threshold. The minimum action for gravitational coupling is $J_{LK} = (\sigma_{\pi N}/\Lambda_{QCD}) \cdot \hbar = \hbar/4$, where $\sigma_{\pi N} = 50$ MeV is the pion-nucleon sigma term and $\Lambda_{QCD} = 200$ MeV is the QCD scale, both measured experimentally. This threshold classifies all known particles: the Z boson, W boson, and top quark ($\Delta t \sim 10^{-25}$ s) do not gravitate; the $\Delta(1232)$ and ρ meson lie at the threshold; protons, neutrons, and electrons gravitate at full strength. Virtual vacuum fluctuations do not gravitate because (i) they are spatially homogeneous and (ii) individually short-lived. The gravitational coupling operates through two stiffness scales — cosmological (R^2 -term, Starobinsky) and local (QCD condensate) — neither of which involves Planck units. We show that the Planck length is a virtual focal point of dimensional analysis: the three constants $\hbar$, G , c converge there mathematically but no physical process occurs at that scale. The physical threshold of gravitational coupling lies twenty orders of magnitude higher, at the QCD scale, and is fully determined by measured nuclear physics quantities. We do not claim that quantum gravity effects are absent — only that the gravitational activation threshold is set by QCD, not by Planck units.

Keywords: gravitational activation, chiral condensate, cosmological constant problem, QCD vacuum, Planck scale, spacetime stiffness

1. Introduction

The cosmological constant problem — the 10^{120} -fold discrepancy between the theoretical vacuum energy density and the observed value — has persisted for nearly six decades since Zel'dovich's 1967 identification of quantum vacuum energy with Einstein's cosmological constant [1]. The present work addresses a logically prior question: under what conditions does energy gravitate at all?

The standard assumption, inherited from Zel'dovich, is that all energy gravitates — that the full energy-momentum tensor of every quantum field contributes to spacetime curvature through Einstein's equations. This assumption was never derived; it was postulated by analogy with the tensor structure of the vacuum expectation value $\langle T^{\mu\nu} \rangle \propto g^{\mu\nu}$ [2]. Five independent theoretical frameworks have demonstrated that this identification is not mandatory: trace-free gravity [3], Kaloper-Padilla sequestering [4], Volovik's analogy with condensed matter [5], Solà Peracaula's running vacuum models [6], and the QKDE framework [7].

In a series of publications [8–14], the present author developed a model of the gravitating vacuum in which the equilibrium quantum vacuum does not gravitate. Only the shift of the chiral condensate $\langle \bar{q}q \rangle$ in the presence of baryonic matter — measured by the pion-nucleon sigma term $\sigma_{\pi N} \approx 50$ MeV [15,16] — produces a Lorentz-scalar perturbation with equation of state $w = -1$ that couples to gravity. The predicted cosmological constant $\Lambda_0 = \alpha \cdot f_b \cdot \sigma^6 / m_{Pl}^4$ yields $1.8 \times 10^{-52} \text{ m}^{-2}$, compared with the observed $1.1 \times 10^{-52} \text{ m}^{-2}$ — an improvement of 119 orders of magnitude over Zel'dovich's estimate [8].

The present paper extends this programme by deriving a gravitational activation threshold: the minimum action required for a localized energy perturbation to produce spacetime curvature. This threshold emerges from two physical mechanisms — the spatial homogeneity of the equilibrium vacuum and the finite response time of the chiral condensate — and is expressed entirely through measured QCD quantities, without reference to Planck units.

The paper is organised as follows. Section 2 establishes the two mechanisms of vacuum non-gravitation. Section 3 derives the gravitational activation threshold and identifies the condensate mass parameter with Λ_{QCD} . Section 4 presents the Lagrangian formulation. Section 5 introduces the two stiffness scales of spacetime. Section 6 classifies particles by their gravitational activation status, including uncertainty analysis. Section 7 reinterprets Planck units within this framework. Section 8 presents the self-consistent cycle, the uniqueness argument, and the determination of $\hbar$ as an eigenvalue of the cycle. Section 9 identifies falsifiable predictions. Section 10 concludes.

2. Two Mechanisms of Vacuum Non-Gravitation

2.1 Spatial homogeneity

Einstein's equations relate spacetime curvature to the energy-momentum tensor:

$$G^{\mu\nu} + \Lambda_0 g^{\mu\nu} = \frac{8\pi G}{c^4} T^{\mu\nu}$$

In the trace-free formulation developed by Ellis [3], the cosmological constant Λ_0 is a constant of integration on the geometric side of the equation, not a manifestation of vacuum energy. The equilibrium vacuum energy ρ_{vac} does not appear on the right-hand side.

The physical basis for this separation is homogeneity. Consider the analogy of a uniformly loaded elastic membrane (trampoline). A uniform load — however large — produces no local curvature, because there is no pressure gradient. The membrane deflects uniformly but does not bend. Only a non-uniform load creates local curvature.

The quantum vacuum is the analogue of the uniform load. Virtual particle pairs appear at all spacetime points with equal probability density. The energy density is enormous ($\sim \Lambda_{QCD}^4/(\hbar c)^3 \sim 10^{36} \text{ J/m}^3$) but spatially homogeneous. In the trace-free formulation, this homogeneous energy is absorbed into the geometric side of the equation — it contributes to Λ_0 , not to $T^{\mu\nu}$.

This is not a regularisation trick. It is a physical statement: homogeneous energy does not gravitate locally. Only inhomogeneities — deviations from the equilibrium vacuum — produce curvature.

2.2 Finite response time of the chiral condensate

Even local, inhomogeneous energy perturbations do not automatically gravitate. The gravitational coupling of matter to geometry, in this framework, is mediated by the chiral condensate. The mechanism is:

1. Baryonic matter is present at a point in spacetime.
2. The presence of baryonic matter shifts the chiral condensate $\langle \bar{q}q \rangle$ from its equilibrium value.
3. The shifted condensate has a Lorentz-scalar energy density with $T^{\mu\nu} \propto g^{\mu\nu}$, i.e., equation of state $w = -1$. This was confirmed independently by Cohen [17].
4. This shifted condensate couples to gravity and produces local curvature.

The condensate is not infinitely responsive. It is a QCD object whose dynamics are governed by the confinement scale Λ_{QCD} .

3. The Gravitational Activation Threshold

3.1 Identification of the condensate mass parameter

The condensate shift field $\phi(x) = \langle \bar{q}q \rangle(x) - \langle \bar{q}q \rangle_0$ satisfies a Klein-Gordon equation (derived in Section 4). The mass parameter μ of this equation is identified with $\Lambda_{QCD}/(\hbar c)$ on the following grounds.

The chiral condensate is an order parameter of QCD. Its fluctuations are the pseudo-Goldstone bosons — the pions. The pion mass $m_\pi = 135 \text{ MeV}$ would be zero in the chiral limit (massless quarks); it is nonzero because of explicit chiral symmetry breaking by quark masses. The natural mass scale for the condensate dynamics, in the absence of fine-tuning, is not m_π but Λ_{QCD} — the scale at which the QCD coupling becomes strong and confinement sets in.

This identification is standard in chiral perturbation theory: Λ_{QCD} sets the radius of convergence of the chiral expansion, the scale of higher-resonance contributions, and the inverse confinement radius [18]. The condensate response time is therefore:

$$\tau_{LK} = \frac{1}{\mu c} = \frac{\hbar}{\Lambda_{QCD}} \approx \frac{1.055 \times 10^{-34} \text{ J}\cdot\text{s}}{3.204 \times 10^{-11} \text{ J}} = 3.29 \times 10^{-24} \text{ s}$$

This is the time required for the gluon field configuration to reorganise in response to a colour source — the inverse confinement scale expressed as a time.

3.2 Derivation of the threshold

For a baryonic perturbation of duration Δt , the condensate shift amplitude evolves as:

$$\phi(t) = \phi_{eq} [1 - e^{-t/\tau_{LK}}]$$

where $\phi_{eq} = \sigma_{\pi N} = 50 \text{ MeV}$ is the equilibrium shift for a static nucleon, measured in pion-nucleon scattering [15,16].

The gravitational action — the product of the condensate shift energy and the duration — is:

$$S_{grav} = \sigma_{\pi N} \cdot [1 - e^{-\Delta t/\tau_{LK}}] \cdot \Delta t$$

The threshold is defined as $S_{grav} = J_{LK}$ at $\Delta t = \tau_{LK}$:

$$J_{LK} = \sigma_{\pi N} \cdot (1 - e^{-1}) \cdot \tau_{LK} \approx 0.632 \cdot \sigma_{\pi N} \cdot \tau_{LK}$$

For the characteristic threshold we use the product $\sigma_{\pi N} \cdot \tau_{LK}$:

$$J_{LK} = \frac{\sigma_{\pi N}}{\Lambda_{QCD}} \cdot \hbar = \frac{50 \text{ MeV}}{200 \text{ MeV}} \cdot \hbar = \frac{\hbar}{4} = 2.64 \times 10^{-35} \text{ J}\cdot\text{s}$$

3.3 The nature of the threshold: quantum or classical?

The constant $\hbar$ enters this expression through the dimensional relationship $\tau_{LK} = \hbar/\Lambda_{QCD}$ — i.e., through the conversion between energy and inverse time that is fundamental to quantum mechanics. The threshold is quantum-mechanical in origin: it arises because the condensate is a quantum field whose response time is set by the uncertainty relation $\Delta E \cdot \Delta t \sim \hbar$.

However, the threshold is not a Bohr-Sommerfeld quantisation condition. There is no quantum number n such that $S = n\hbar/4$. Rather, the threshold is a classical activation condition imposed by quantum dynamics: the condensate (a quantum field) has a response time (a quantum timescale), and perturbations shorter than this timescale do not activate the gravitational coupling. The situation is analogous to the photoelectric effect: the threshold frequency $\nu_0 = W/h$ is a classical cutoff imposed by quantum mechanics.

The physically meaningful content is the dimensionless ratio $\sigma_{\pi N}/\Lambda_{QCD} = 1/4$, which is a property of QCD dynamics computable from first principles in lattice QCD.

3.4 Uncertainty analysis

The QCD scale Λ_{QCD} depends on the renormalisation scheme and the number of active flavours. In the $\overline{MS}$ scheme with $n_f = 3$, current determinations give $\Lambda_{QCD} = 200\text{--}340$ MeV [19]. The sigma term is determined as $\sigma_{\pi N} = 40\text{--}60$ MeV from lattice QCD [16] and pion-nucleon scattering analysis [15].

The threshold ratio $\sigma_{\pi N}/\Lambda_{QCD}$ therefore has an uncertainty:

$$\frac{\sigma_{\pi N}}{\Lambda_{QCD}} = 0.15\text{--}0.30 \quad (\text{central value } 0.25)$$

This propagates to the activation function. For the ρ meson ($\tau = 4.5 \times 10^{-24}$ s):

Λ_{QCD} (MeV)	τ_{LK} (10^{-24} s)	$\Delta t/\tau_{LK}$	f (%)
200	3.29	1.37	65
250	2.63	1.71	74
340	1.94	2.32	84

The qualitative classification is robust: the ρ meson is always in the threshold region (50–85%), and the Z boson is always far below threshold ($f < 1\%$ for all Λ_{QCD}). The boundary between “does not gravitate” and “gravitates” shifts within the hadronic resonance region but never approaches the Planck scale.

4. Lagrangian Formulation

The complete action consists of three terms:

$$S = S_{grav} + S_{cond} + S_{int}$$

4.1 Gravitational sector

$$S_{grav} = \int \left[\frac{R}{16\pi G} + \frac{R^2}{6M^2} \right] \sqrt{-g} d^4x$$

The first term is the standard Einstein-Hilbert action. The second is the Starobinsky R^2 -term [20], where $M \approx 3 \times 10^{13}$ GeV is the scalaron mass determined from Planck satellite CMB data [21]. This term provides the cosmological stiffness of spacetime (Section 5).

4.2 Condensate sector

$$S_{cond} = \int \left[\frac{1}{2} (\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2} \mu^2 \phi^2 \right] \sqrt{-g} d^4x$$

Here $\phi(x) = \langle \bar{q}q \rangle(x) - \langle \bar{q}q \rangle_0$ is the shift of the chiral condensate from equilibrium. The mass parameter is $\mu = \Lambda_{QCD}/(\hbar c)$, identified with the inverse confinement radius as argued in Section 3.1. The field ϕ is the mediator between matter and geometry.

4.3 Matter-condensate coupling

$$S_{int} = \int \alpha \cdot \phi \cdot n_B \sqrt{-g} d^4x$$

where n_B is the baryon number density and $\alpha = \sigma_{\pi N}/m_N \approx 0.005$ is the coupling constant derived from sigma terms without cosmological fitting [8,9].

4.4 Equation of motion for the condensate

Variation with respect to ϕ yields:

$$(\square + \mu^2)\phi(x) = \alpha \cdot n_B(x) \cdot m_N$$

This is a Klein-Gordon equation with source. The mass $\mu = \Lambda_{QCD}/(\hbar c)$ determines both the spatial range $r_{LK} = 1/\mu = \hbar c/\Lambda_{QCD} = 0.99$ fm and the temporal response $\tau_{LK} = 1/(\mu c) = \hbar/\Lambda_{QCD} = 3.29 \times 10^{-24}$ s.

For a static point-like nucleon, the solution is a Yukawa potential:

$$\phi(r) = \frac{\alpha \cdot m_N}{4\pi r} e^{-\mu r} = \frac{\sigma_{\pi N}}{4\pi r} e^{-r/r_{LK}}$$

For an impulsive source (a particle with lifetime Δt), the temporal envelope is:

$$\phi_{max}(\Delta t) = \frac{\sigma_{\pi N}}{4\pi} [1 - e^{-\Delta t/\tau_{LK}}]$$

The condensate shift ϕ is a Lorentz scalar. As shown by Cohen, Furnstahl, and Griegel [17], the in-medium chiral condensate transforms as a scalar under Lorentz boosts, so its energy-momentum tensor takes the form $T_{vac}^{\mu\nu} = -\phi(x) \cdot g^{\mu\nu}$, corresponding to equation of state $w = -1$. This local, matter-induced vacuum energy enters Einstein's equations as a source of curvature.

5. Two Stiffness Scales

Spacetime possesses two distinct stiffness scales, analogous to a physical membrane that has both a bulk tension (resistance to overall stretching) and a bending modulus (resistance to local deformation).

5.1 Cosmological stiffness

The R^2 -term in the gravitational action, with coefficient $1/(6M^2)$ where $M \approx 3 \times 10^{13}$ GeV, governs the large-scale relaxation of the universe toward its asymptotic de Sitter state [13]. The associated response time is:

$$\tau_{cosmo} = \frac{\hbar}{Mc^2} \approx 2.2 \times 10^{-38} \text{ s}$$

This stiffness determines how the universe relaxes to Λ_0 and governs inflationary dynamics.

5.2 Local stiffness (QCD condensate)

The condensate response time $\tau_{LK} = \hbar/\Lambda_{QCD} \approx 3.3 \times 10^{-24}$ s governs the local coupling of matter to geometry. This stiffness determines which particles gravitate and which do not. It operates on the scale of 1 fm.

5.3 Comparison

The two stiffnesses differ by fourteen orders of magnitude:

$$\frac{\tau_{LK}}{\tau_{cosmo}} = \frac{Mc^2}{\Lambda_{QCD}} \approx 1.5 \times 10^{14}$$

They govern different physical processes at different scales and enter the action through different terms. The cosmological stiffness is a property of the gravitational sector (R^2 -term); the local stiffness is a property of the matter-geometry coupling (condensate dynamics).

6. Classification of Particles

6.1 Activation function

The fraction of full gravitational coupling achieved by a particle with lifetime Δt is:

$$f\left(\frac{\Delta t}{\tau_{LK}}\right) = [1 - e^{-\Delta t/\tau_{LK}}]^2$$

This function approaches 0 for $\Delta t \ll \tau_{LK}$ and 1 for $\Delta t \gg \tau_{LK}$.

6.2 Results

Table 1 classifies known particles by their gravitational activation status, using $\Lambda_{QCD} = 200$ MeV (with uncertainty range for $\Lambda_{QCD} = 200\text{--}340$ MeV in parentheses where the classification is sensitive).

Table 1. Gravitational activation of particles

Particle	Mass	Lifetime Δt	$\Delta t/\tau_{LK}$	f (%)	Status
Z boson	91.2 GeV	3×10^{-25} s	0.091	0.8	Does not gravitate
W boson	80.4 GeV	3×10^{-25} s	0.091	0.8	Does not gravitate
Top quark	173 GeV	5×10^{-25} s	0.15	2.1	Does not gravitate
ρ meson	775 MeV	4.5×10^{-24} s	1.37	49 (55–72)	Threshold
$\Delta(1232)$	1232 MeV	5.6×10^{-24} s	1.70	56 (62–77)	Threshold
ω meson	783 MeV	7.7×10^{-23} s	23.4	99.8	Gravitates

Particle	Mass	Lifetime Δt	$\Delta t/\tau_{LK}$	f (%)	Status
π^0	135 MeV	8.4×10^{-17} s	2.6×10^7	100	Gravitates
Muon	105.7 MeV	2.2×10^{-6} s	6.7×10^{17}	100	Gravitates
Neutron	939.6 MeV	879 s	2.7×10^{26}	100	Gravitates
Proton	938.3 MeV	stable	∞	100	Gravitates
Electron	0.511 MeV	stable	∞	100	Gravitates

The threshold passes through the hadronic resonance region (10^{-24} – 10^{-23} s). This is the QCD scale, not the Planck scale.

6.3 Virtual particles

For virtual particles, $\Delta t = \hbar/E$. At energies $E > \Lambda_{QCD}$, the lifetime $\Delta t < \tau_{LK}$, and the condensate does not respond. At energies $E < \Lambda_{QCD}$, the lifetime exceeds τ_{LK} , but the particles are part of the equilibrium vacuum: they appear at all spacetime points with equal probability density, and the first mechanism (Section 2.1) applies. In both regimes, virtual particles do not gravitate.

7. The Planck Scale as Virtual Focal Point

7.1 Dimensional analysis revisited

The Planck length $\ell_P = \sqrt{\hbar G/c^3} \approx 1.6 \times 10^{-35}$ m is commonly interpreted as the scale at which spacetime structure changes. This interpretation extrapolates G — a macroscopic quantity determined by the collective vacuum structure (Sakharov [22]) — to scales 10^{20} times smaller than the QCD confinement radius.

7.2 The lens analogy

A diverging lens produces a virtual focal point: the point from which diverging rays appear to originate, if extrapolated backward. No light converges there. No physical process occurs there. The virtual focus is a mathematical construct.

The Planck scale is the virtual focal point of three constants. $\hbar$ and c are properties of the spacetime structure ($\hbar$ is the grain of phase space; c is the propagation speed, a stiffness property of the membrane). G is the induced stiffness (Sakharov: $G^{-1} \sim N\Lambda_{UV}^2$). Combining all three and solving for a length yields ℓ_P — a virtual intersection. The constants converge there mathematically. No physical process occurs there.

7.3 Planck units derived from LK units

$$\ell_P = r_{LK} \times \frac{\Lambda_{QCD}}{E_{Pl}}, \quad t_P = \tau_{LK} \times \frac{\Lambda_{QCD}}{E_{Pl}}$$

where $E_{Pl} = \sqrt{\hbar c^5/G}$. The ratio $\Lambda_{QCD}/E_{Pl} = 1.64 \times 10^{-20}$ contains G and nothing else. In Sakharov's framework, G is induced by the vacuum structure, so G is secondary to QCD.

The hierarchy is: QCD (measured) $\rightarrow G$ (induced) $\rightarrow$ Planck (derived). Planck units are a shadow of QCD, projected through G .

8. Self-Consistent Cycle and Uniqueness

8.1 The cycle

The gravitating vacuum model defines a cyclic operator Φ_{cycle} [8]:

1. **Vacuum assignment:** the condensate distribution $\phi(x)$ determines local vacuum energy.
2. **Metric solution:** Einstein's equations (with R^2 -term) yield the spacetime geometry.
3. **Matter redistribution:** matter responds to geometry; baryon density $\rho_B(x)$ adjusts.
4. **Return to step 1:** adjusted ρ_B determines a new condensate shift.

8.2 Lipschitz constant

The Lipschitz constant k is estimated from cosmological N -body simulations (resolution 256^3) by running the cycle from two different initial density fields and comparing the L^2 -norm of the resulting density fields at each iteration [11]. The ratio of output distance to input distance gives $k \approx 0.004 \pm 0.002$. The method is detailed in [11]; the key result is that k is determined by the smallness of $\alpha = 0.005$ (the vacuum-matter coupling) multiplied by the baryon fraction $f_b = 0.156$, giving $k \sim \alpha \cdot f_b \cdot (\text{geometric factor}) \approx 0.004$.

8.3 Fixed-point theorem and its scope

Since $k < 1$, the Banach fixed-point theorem guarantees that the cycle has exactly one fixed point — one self-consistent configuration of vacuum, geometry, and matter — and converges to this fixed point from any initial conditions within the domain of the operator.

An important qualification: the Banach theorem guarantees uniqueness of the fixed point *for a given operator* Φ_{cycle} . The operator itself is defined in terms of the physical laws (Einstein's equations, QCD, the condensate coupling). The theorem does not, by itself, prove that the fundamental constants are determined by the cycle. What it does prove is that once the laws are given, the solution — the configuration of the universe — is unique.

8.4 The constants as a self-consistent package

The stronger claim — that the constants themselves are mutually determined — requires an additional argument, which we present as a hypothesis with supporting evidence.

The vacuum is a quantum system quantised in units of $\hbar$. This quantisation determines the number of accessible states, which determines the structure of the vacuum: the chiral condensate, Λ_{QCD} , $\sigma_{\pi N}$. The vacuum structure, through Sakharov's mechanism [22], determines G : the gravitational constant is induced by quantum fields, with $G^{-1} \sim N\Lambda_{UV}^2$, where both N and Λ_{UV} depend on $\hbar$.

G , combined with $\hbar$ and c , yields the Planck scale. But the Planck scale must be coherent with the same $\hbar$ that quantised the vacuum in the first step. If one attempts to quantise the

vacuum in units of $2\hbar$, the resulting vacuum has different Λ_{QCD} , different $\sigma_{\pi N}$, different G — and when the cycle returns to the beginning, it requires $\hbar$, not $2\hbar$, for self-consistency.

Only one value of the quantum of action produces a vacuum that, when quantised by that same quantum, reproduces itself. $\hbar$ is not a free parameter — it is the eigenvalue of the self-consistent cycle [23]. The same argument applies to c (a property of the membrane whose stiffness is determined by the vacuum) and to G (induced by the vacuum). All constants are determined simultaneously as a single fixed point.

This claim is falsifiable: if a self-consistent solution exists for a different value of $\sigma_{\pi N}/\Lambda_{QCD}$, the uniqueness claim fails. Current evidence — $k \approx 0.004$, a single fixed point in all simulations — supports uniqueness but does not constitute a proof.

8.5 Connection to Leibniz

This result connects to Leibniz’s concept of *optimum* — not “morally best” (a distortion introduced by Voltaire) but “maximally coherent under constraints,” from the Latin *ops* (capability) [24]. The fixed point of the self-consistent cycle is the Leibnizian optimum made precise: the unique configuration that satisfies all constraints simultaneously. A universe with different constants is not “another possible universe” but a configuration that does not close the cycle — a non-fixed-point, and therefore not persistent.

9. Falsifiable Predictions

A1. Partial gravitational activation of hadronic resonances. The $\Delta(1232)$ and $\rho(770)$ gravitate at 50–75% (depending on Λ_{QCD}). Testable via lattice QCD computation of the gravitational form factor in the trace-free formulation.

A2. No Planck-scale discreteness. No experiment will detect minimum length, modified dispersion relations, or spacetime discreteness at ℓ_P . Existing null results from Fermi observations [25] are consistent; future tests should continue to yield null results.

A3. Sharp truncation of rotation curves at the vacuum capture radius r_c , distinguishable from NFW profiles. Testable with SKA and ngVLA [12].

A4. Baryon-dependent satellite dark mass. M_{dark}/M_{baryon} of satellite galaxies should correlate with distance to the host center. Testable with DESI and Euclid [12].

B1. Lattice QCD verification that $\sigma_{\pi N}/\Lambda_{QCD}$ stabilises near $1/4$ with controlled systematics.

B2. Convergence of the cyclic operator. N -body simulations at $\geq 512^3$ should confirm $k < 0.01$.

C1. No second fixed point. Proof that the Banach contraction holds globally would establish uniqueness as theorem.

10. Conclusion

We have derived a gravitational activation threshold $J_{LK} = (\sigma_{\pi N}/\Lambda_{QCD}) \cdot \hbar = \hbar/4$ from QCD condensate dynamics. This threshold classifies all known particles into those that gravitate and those that do not, with the boundary at the QCD scale — twenty orders of magnitude above the Planck scale. The 10^{120} cosmological constant problem is not resolved by finding the correct calculation of vacuum energy — it is dissolved by recognising that most of the vacuum does not gravitate.

The Planck length is not a physical boundary but a virtual focal point of dimensional analysis, analogous to the virtual focus of a diverging lens. The physical threshold of gravitational coupling is set by a measurable ratio: $\sigma_{\pi N}/\Lambda_{QCD} = 1/4$.

Statements and Declarations

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Use of AI tools: Computational verification of numerical results (Table 1 and dimensional analysis) was performed with the assistance of Claude (Anthropic). All physical arguments, derivations, and interpretations are the author's own.

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